

BLACK HILLS, SD, USA

WMO: 726605

Lat: 44.483N

Lon: 103.783W

Elev: 1198

StdP: 87.73

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 00386

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-22.3	-19.1	-26.2	0.4	-21.1	-22.9	0.6	-18.4	14.5	8.3	12.6	6.8	2.8	300	0.628

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.0	32.6	18.5	30.9	18.3	28.9	18.2	21.5	27.8	20.4	27.0	19.4	26.5	4.7	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.1	16.1	25.9	17.9	14.9	24.1	17.3	14.3	23.2	68.8	27.9	64.5	26.7	60.9	26.6	25.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.0	10.8	9.2	DB	-26.0	35.6	2.1	2.3	-27.5	37.2	-28.8	38.6	-30.0	39.8	-31.5	41.5
			WB	-26.2	23.6	2.0	1.3	-27.6	24.5	-28.8	25.2	-29.9	25.9	-31.4	26.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.9	-2.7	-4.2	2.4	6.1	11.0	17.2	21.4	20.0	16.1	7.9	2.3	-3.6		
	DBStd	10.65	7.12	8.06	6.98	5.56	4.84	3.98	3.30	3.41	4.91	5.89	6.67	7.70		
	HDD10.0	2002	395	398	246	138	46	2	0	0	10	108	238	421		
	HDD18.3	4052	653	630	493	368	233	66	8	22	97	325	479	678		
	CDD10.0	1239	1	2	11	20	76	218	355	311	193	43	8	1		
	CDD18.3	247	0	0	0	1	5	31	104	74	29	2	0	0		
	CDH23.3	2956	0	0	2	12	61	439	1158	853	396	35	0	0		
	CDH26.7	1036	0	0	0	1	9	136	452	312	119	6	0	0		
(14) Wind	WSAvg	4.1	4.7	4.2	4.1	4.5	4.1	3.9	3.5	3.4	3.5	4.2	4.2	4.3		
(15) Precipitation	PrecAvg	643	28	31	42	71	102	97	65	49	44	58	32	25		
	PrecMax	916	62	55	85	159	251	212	143	100	128	181	100	61		
	PrecMin	444	11	11	24	22	32	18	18	16	18	3	3	3		
	PrecStd	131	13	11	15	31	57	54	34	26	28	49	22	17		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.2	17.7	22.8	25.0	27.9	33.0	35.9	34.1	32.4	27.4	21.2	17.0		
		MCWB	4.9	7.3	10.3	12.2	17.3	19.1	18.3	17.4	17.3	14.2	9.6	7.0		
	2%	DB	12.0	13.7	19.0	21.4	24.9	30.0	32.9	32.2	29.9	23.8	17.8	12.3		
		MCWB	4.6	5.9	8.4	11.2	15.2	18.2	19.0	18.3	17.0	11.9	8.1	4.5		
	5%	DB	9.1	10.1	16.2	17.9	22.6	27.8	31.2	29.9	27.6	20.3	15.0	8.9		
		MCWB	3.2	3.8	6.7	9.5	14.5	17.9	18.9	18.1	15.9	10.8	6.7	2.9		
	10%	DB	7.4	7.3	12.8	15.3	20.0	25.9	29.0	27.9	25.2	17.6	12.4	7.1		
		MCWB	2.1	2.1	5.6	8.0	13.2	17.3	19.0	17.5	14.7	9.6	5.2	1.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.0	7.8	10.9	13.3	19.7	22.2	23.9	21.8	20.6	15.0	10.2	7.4		
		MCDB	13.4	16.0	21.5	22.7	26.3	28.4	29.9	28.2	28.7	24.6	19.8	16.3		
	2%	WB	4.7	5.8	8.7	11.5	16.8	20.5	22.0	20.6	18.7	13.2	8.6	4.8		
		MCDB	10.7	13.2	17.7	19.7	23.0	26.7	28.1	27.7	26.5	21.3	17.2	11.0		
	5%	WB	3.3	4.0	7.3	10.2	15.2	19.1	20.8	19.7	17.2	11.6	7.1	3.1		
		MCDB	8.6	9.9	15.4	17.4	20.9	25.7	26.8	26.7	25.2	19.4	14.9	8.4		
	10%	WB	2.2	2.1	5.9	8.8	13.6	17.8	19.8	18.6	15.7	10.1	5.5	1.8		
		MCDB	7.1	6.6	12.4	15.3	18.9	24.3	26.9	26.0	23.5	16.6	11.6	6.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.0	10.0	11.0	11.3	10.7	12.2	13.0	13.1	13.2	10.8	10.6	9.2		
		MCDBR	12.7	13.4	16.0	16.4	14.4	16.1	15.4	15.8	16.5	15.4	14.9	11.9		
		MCWBR	7.8	7.6	8.5	9.0	8.2	8.3	6.5	7.4	7.7	7.9	7.9	7.2		
	5% WB	MCDBR	12.2	12.9	15.2	15.3	13.1	14.6	13.1	13.8	14.7	14.6	14.4	11.3		
		MCWBR	8.0	7.6	8.4	8.8	8.0	8.8	8.0	7.9	7.7	7.9	7.8	7.3		
(40) Clear-Sky Solar Irradiance	taub		0.253	0.260	0.277	0.306	0.320	0.334	0.354	0.356	0.320	0.289	0.262	0.249		
	taud		2.583	2.555	2.561	2.472	2.503	2.486	2.460	2.432	2.520	2.581	2.570	2.578		
	Ebn at Noon		902	952	973	960	949	933	909	896	908	902	879	872		
	Edh at Noon		62	76	87	103	104	106	108	107	90	74	62	56		
(44) All-Sky Solar Radiation	RadAvg		1.86	2.83	4.01	4.95	5.89	6.87	7.09	6.10	4.75	3.09	2.06	1.55		
	RadStd		0.09	0.16	0.31	0.28	0.59	0.56	0.26	0.24	0.31	0.29	0.17	0.11		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page